

Clinical Risk Assessment Report: Patient 15

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 9.3% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.42x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.6%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -2.06 [Bottom 3%] (-2.6%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: 0.37 [Top 10%] (-1.1%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 0.95 [Top 16%] (-0.8%)**

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.43 [Bottom <1%] (-0.7%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.77 [Bottom 4%] (-0.7%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: 0.18 [Top 8%] (-0.6%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.9 [Avg (74th %ile)] (-0.6%)**

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: -2.4 [Bottom 3%] (-0.6%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Machine Learning Feature Attribution (SHAP Decision Path)

